Elham Aghakhani

Philadelphia, PA

elham.aghakhani@drexel.edu

Personal Website Linkedin

Google Scholar Github

Education

- **PhD in Information Science** — Jan 2024 – Expected June 2028
  Drexel University

- **MSc in Computer Science, Artificial Intelligence** — Oct 2020 – Sep 2023
  K. N. Toosi University of Technology
  - **Coursework:** Machine Learning, Pattern Recognition, Big Data Analysis, Reinforcement Learning
  - **Thesis:** Automatic Persian text summarization using pre-trained language models and reinforcement learning

- **BSc in Computer Science** — Sep 2015 – Sep 2019
  K. N. Toosi University of Technology
  - **Thesis:** Implementing an automatic system to collect news from news agencies

Research Interests

- LLM Evaluation
- Human-centered NLP
- Behavioral modeling
- Computational Social Science and Sociotechnical Systems

Publications

- **Aghakhani, E.**, & Rezapour, R. (2026). Like a Therapist, But Not. Reddit Narratives of AI in Mental Health Contexts. *Findings of the Association for Computational Linguistics (ACL 2026)*.

- **Aghakhani, E.**, Wang, L., Washington, K. T., Demiris, G., Huh-Yoo, J., & Rezapour, R. (2025). From Conversation to Automation. Leveraging LLMs for Problem-Solving Therapy Analysis. *Findings of the Association for Computational Linguistics (ACL 2025)*, 25189–25207, Vienna, Austria.

- Bouzoubaa, L., **Aghakhani, E.**, & Rezapour, R. S. (2025). Phenotypes of stigma expressed by people who use drugs on Reddit. *Social Science & Medicine*, 118889.

- Namvarpour, M.*, **Aghakhani, E.***, Ekstrand, M. D., Rezapour, R., & Razi, A. (2025). The evolving landscape of youth online safety. Insights from news media analysis. *WebSci 2025*.

- Wang, L., Lynch, D., **Aghakhani, E.**, Demiris, G., Washington, K., Rezapour, R., & Huh-Yoo, J. (2025). AnnoLoom. Augmenting codebook generation and annotation with large language models. *Proceedings of the Association for Information Science and Technology*.

- Bouzoubaa, L., **Aghakhani, E.**, & Rezapour, R. (2024). Words matter. Reducing stigma in online conversations about substance use with large language models. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)*.

- Bouzoubaa, L., **Aghakhani, E.**, Song, M., Trinh, M., & Rezapour, R. (2024). Decoding the narratives. Analyzing personal drug experiences shared on Reddit. *Findings of the Association for Computational Linguistics (ACL 2024)*.

*Equal contribution.

Experience

Instructor of Record

- **Drexel University**: Introduction to Data Science (INFO103-900) – Summer 2025

Teaching Assistant

- **Drexel University**:
  - Web Systems and Services (INFO151) - Fall 2025
  - Social Collaborative and Computing (INFO616) – Winter 2025
- **K. N. Toosi University of Technology**:
  - Probabilistic Graphical Models – 2021–2022
  - Machine Learning – 2021–2022
  - Statistical Pattern Recognition – 2021–2022

Research Assistant

- **Drexel University** (Jan 2024 – Present)
  - Designed expert-informed annotation rubrics for mental health–related language
  - Designed and analyzed large-scale observational studies using social media and news data.
  - Conducted human-centered analyses of online behavior, stigma, and AI-assisted interventions.
  - Designed prompt-based and simulation-style experiments using LLM agents.
  - Evaluated generative AI systems for social and behavioral outcomes.
  - Used Python for data cleaning, statistical analysis, and modeling.

Industry Experience

- **ERON Company**, Web Developer (Sep 2019 – Mar 2020)
  - Designed and built databases, web applications, and webpages using backend technologies.
- **VADA House of Mobile**, Web Developer (Jan 2019 – Sep 2019)
  - Designed and implemented web applications and APIs using PHP and Laravel.

Peer Review Activities

- Reviewer for: *ACL 2026*, *WebSci'2026*, *ICWSM 2025*, *ACL 2025*, *NAACL 2024*, *EMNLP 2024*

Skills

- **Data Analysis**: Python, Pandas, NumPy, scikit-learn, PyTorch
- **Generative AI and NLP**: LLM fine-tuning, prompt engineering, model evaluation, inference, experiment design, annotation pipelines

Certifications

Natural Language Processing, Coursera (Aug 2021):
- Credential ID: 5WUJGRXXP22V

Machine learning in python with scikit-learn, France Universite Numerique (Jul 2021):
- Credential ID: 4TD52DHM5TCD

Deep Learning workshop, IEEE Iran section (Mar 2021):
- Credential ID: 147474

Awards, Fellowships, and Highlights

Selected to attend the 8th Annual Summer Institute in Computational Social Science, University of Pennsylvania — Philadelphia, PA, July 2024